

Overview of Uncertainty quantification

Subject: Uncertainty quantification

1.

d is the dimension of input variables, and r is the dimension of unknown parameters. While $\mathbf{h}^m(\cdot)$ is pre-defined, $\{\beta_m, \sigma_m, \omega_{km}, k = 1, \dots, d+r\}$, known as hyperparameters of the GP model, need to be estimated via maximum likelihood estimation (MLE). This module can be considered as a generalized kriging method.

2.

Selective methodologies Much research has been done to solve uncertainty quantification problems, though a majority of them deal with uncertainty propagation. During the past one to two decades, a number of approaches for inverse uncertainty quantification problems have also been developed and have proved to be useful for most small- to medium-scale problems.

3.

Simulation-based methods: Monte Carlo simulations, importance sampling, adaptive sampling, etc. General surrogate-based methods: In a non-intrusive approach, a surrogate model is learnt in order to replace the experiment or the simulation with a cheap and fast approximation. Surrogate-based methods can also be employed in a fully Bayesian fashion. This approach has proven particularly powerful when the cost of sampling, e.g. computationally expensive simulations, is prohibitively high. Local expansion-based methods: Taylor series, perturbation method, etc. These methods have advantages when dealing with relatively small input variability and outputs that don't express high nonlinearity. These linear or linearized methods are detailed in the article Uncertainty propagation. Functional expansion-based methods: Neumann expansion, orthogonal or Karhunen–Loeve expansions (KLE), with polynomial chaos expansion (PCE) and wavelet expansions as special cases. Most probable point (MPP)-based methods: first-order reliability method (FORM) and second-order reliability method (SORM). Numerical integration-based methods: Full factorial numerical integration (FFNI) and dimension reduction (DR). For non-probabilistic approaches, interval analysis, Fuzzy theory, Possibility theory and evidence theory are among the most widely used. The probabilistic approach is considered as the most rigorous approach to uncertainty analysis in engineering design due to its consistency with the theory of decision analysis. Its cornerstone is the calculation of probability density functions for sampling statistics. This can be performed rigorously for random variables that are obtainable as transformations of Gaussian variables, leading to exact confidence intervals.

4.

where $y^e(\mathbf{x})$ denotes the experimental measurements as a function of several input variables $\mathbf{x}$, $y^m(\mathbf{x})$ denotes the computer model (mathematical model) response, $\delta(\mathbf{x})$ denotes the additive discrepancy function (aka bias function), and ε denotes the experimental uncertainty. The objective is to estimate the

discrepancy function $\delta(\mathbf{x})$, and as a by-product, the resulting updated model is $y^m(\mathbf{x}) + \delta(\mathbf{x})$. A prediction confidence interval is provided with the updated model as the quantification of the uncertainty.

5.

Full approach Fully Bayesian approach requires that not only the priors for unknown parameters θ but also the priors for the other hyperparameters ϕ should be assigned. It follows the following steps:

6.

Derive the posterior distribution $p(\theta, \phi | \text{data})$; Integrate ϕ out and obtain $p(\theta | \text{data})$. This single step accomplishes the calibration; Prediction of the experimental response and discrepancy function. However, the approach has significant drawbacks:

7.

Dimensionality issue: The computational cost increases dramatically with the dimensionality of the problem, i.e. the number of input variables and/or the number of unknown parameters. Identifiability issue: Multiple combinations of unknown parameters and discrepancy function can yield the same experimental prediction. Hence different values of parameters cannot be distinguished/identified. This issue is circumvented in a Bayesian approach, where such combinations are averaged over. Incomplete model response: Refers to a model not having a solution for some combinations of the input variables. Quantifying uncertainty in the input quantities: Crucial events missing in the available data or critical quantities unidentified to analysts due to, e.g., limitations in existing models. Little consideration of the impact of choices made by analysts.

8.

$R^m((\mathbf{x}, \theta), (\mathbf{x}', \theta')) = \exp\left\{-\sum_{k=1}^d \omega_{km}(\mathbf{x}_k - \mathbf{x}'_k)^2\right\} \exp\left\{-\sum_{k=1}^r \omega_{d+k,m}(\theta_k - \theta'_k)^2\right\}$.

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9.

$\delta(\mathbf{x}) \sim \mathcal{GP}(\mathbf{h}^\delta(\cdot), \mathbf{T}^\delta \beta_\delta, \sigma_\delta^2 \mathbf{R}^\delta(\cdot, \cdot))$

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10.

Mathematical perspective In mathematics, uncertainty is often characterized in terms of a probability distribution. From that perspective, aleatoric uncertainty means that the probability distribution does not unambiguously point to a specific outcome or not being able to predict what a random sample

drawn from a probability distribution will be, while epistemic uncertainty means that one does not know the relevant probability distribution because one lacks a basis for determining it. The figure below shows four different examples of aleatoric and epistemic uncertainty, visualized in two mathematically equivalent ways consisting of a triangle of subjective opinions from subjective logic and as a Beta probability density function.

11.

Uncertainty propagation is the quantification of uncertainties in system output(s) propagated from uncertain inputs. It focuses on the influence on the outputs from the parametric variability listed in the sources of uncertainty. The targets of uncertainty propagation analysis can be:

12.

where $y_m(\mathbf{x}, \boldsymbol{\theta})$ denotes the computer model response that depends on several unknown model parameters $\boldsymbol{\theta}$, and $\boldsymbol{\theta}^*$ denotes the true values of the unknown parameters in the course of experiments. The objective is to either estimate $\boldsymbol{\theta}^*$, or to come up with a probability distribution of $\boldsymbol{\theta}^*$ that encompasses the best knowledge of the true parameter values.